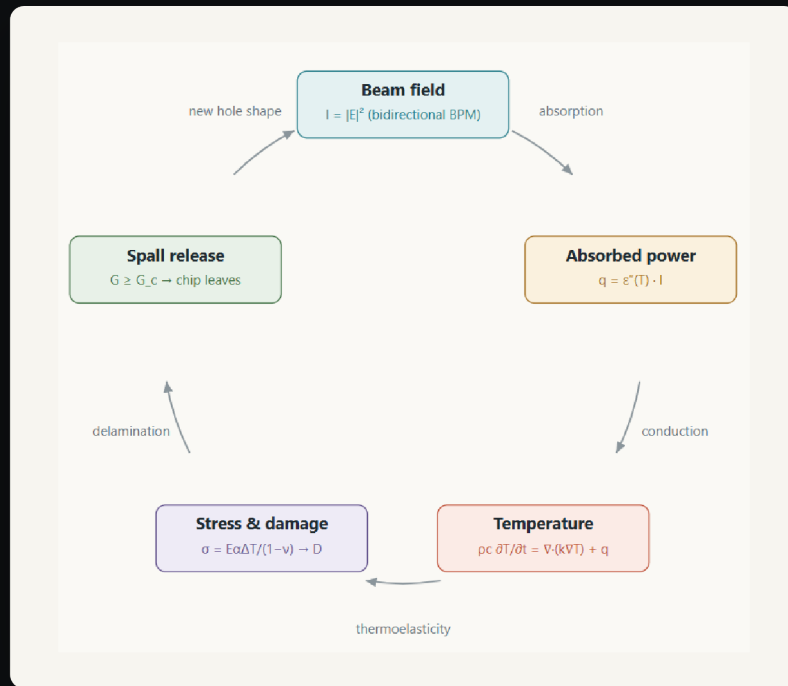


MILLIMETER-WAVE DRILLING PROGRAM

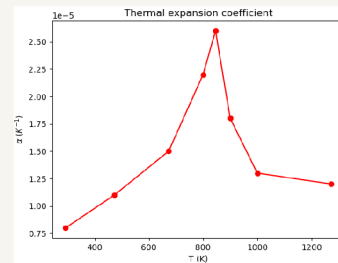
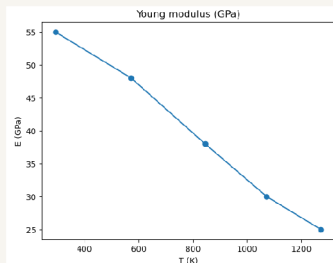
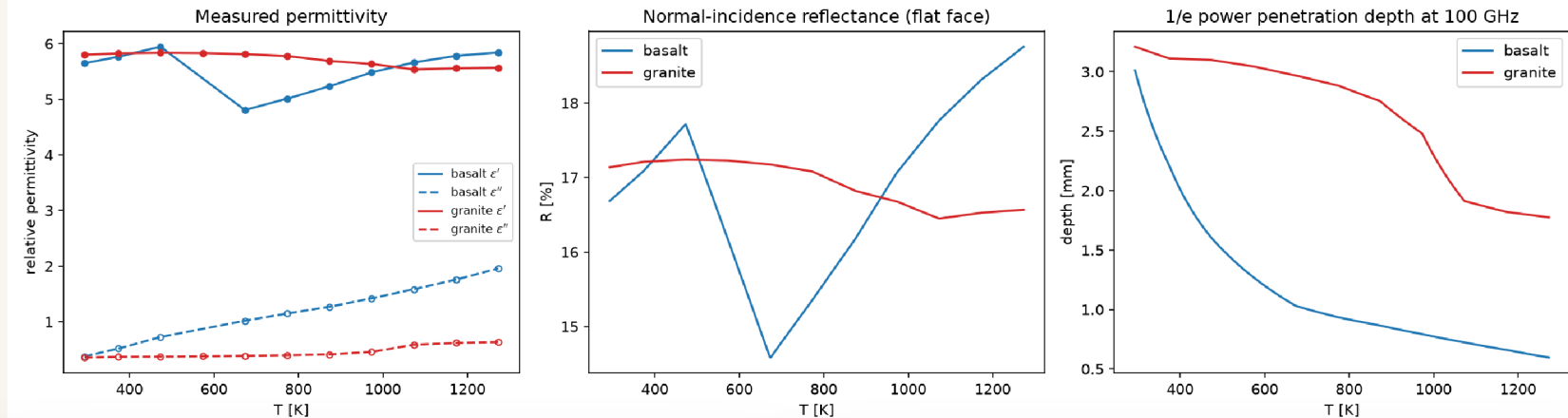
Thermal Spallation Drilling

Physics model, beam-propagation results, and drilling performance

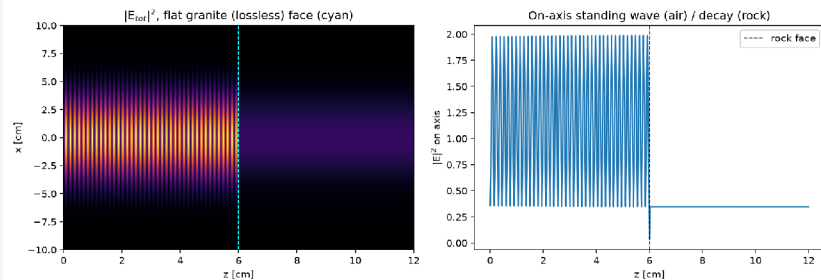
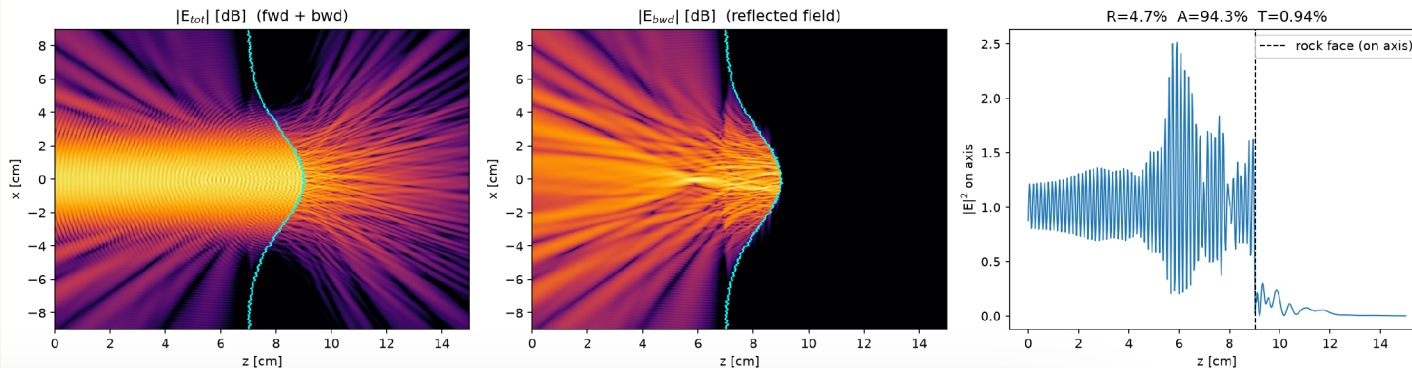
The Millimeter-Wave Spallation Loop



Rock Properties Driving the Physics



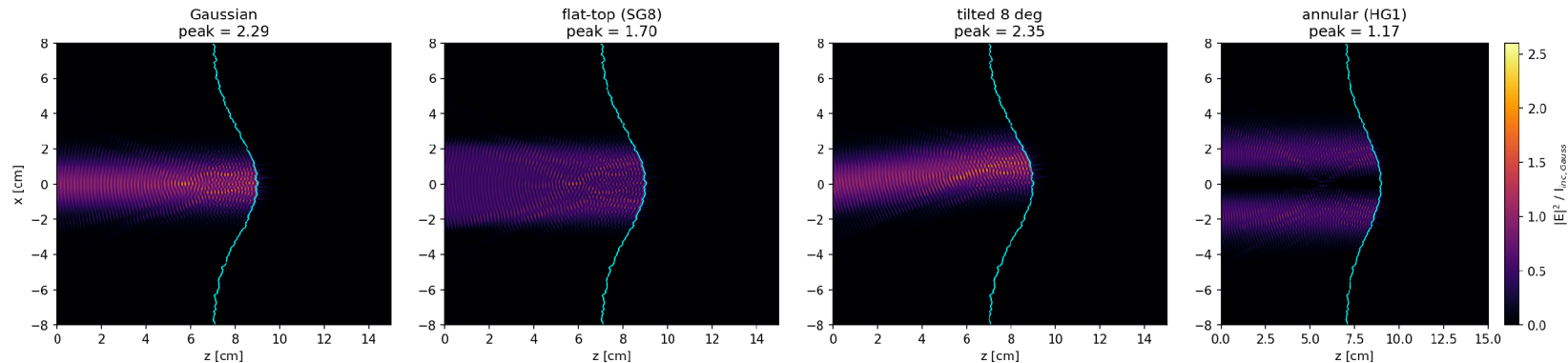
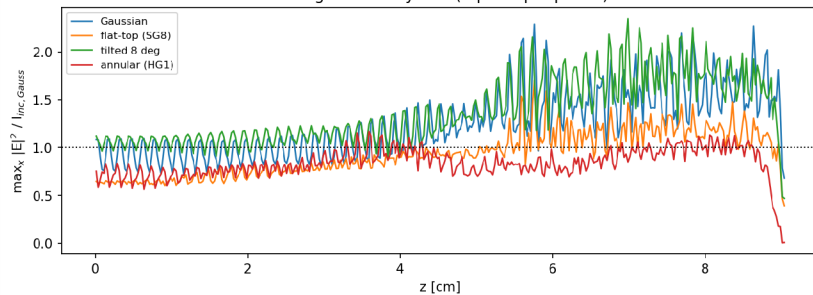
Modeling the Beam–Rock Interaction

100 GHz beam ($w=2$ cm waist) on granite: crater + roughness

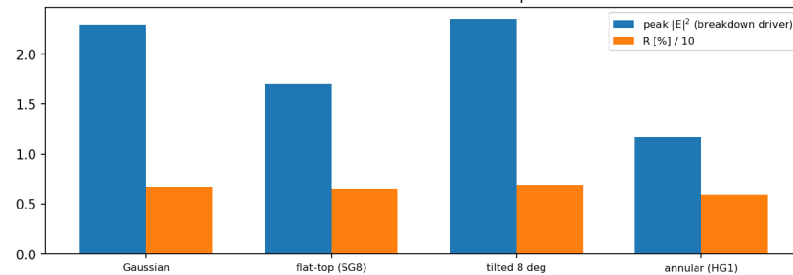
The reflected wave sets up a standing wave pattern in front of the rock face, concentrating intensity in a peak just above the ablation reflection zone before the field decays into the rock.

Choosing a Beam Profile

Beam profile vs standing-wave hot spot: granite, crater + roughness, equal power

Peak gas intensity vs z (equal input power)

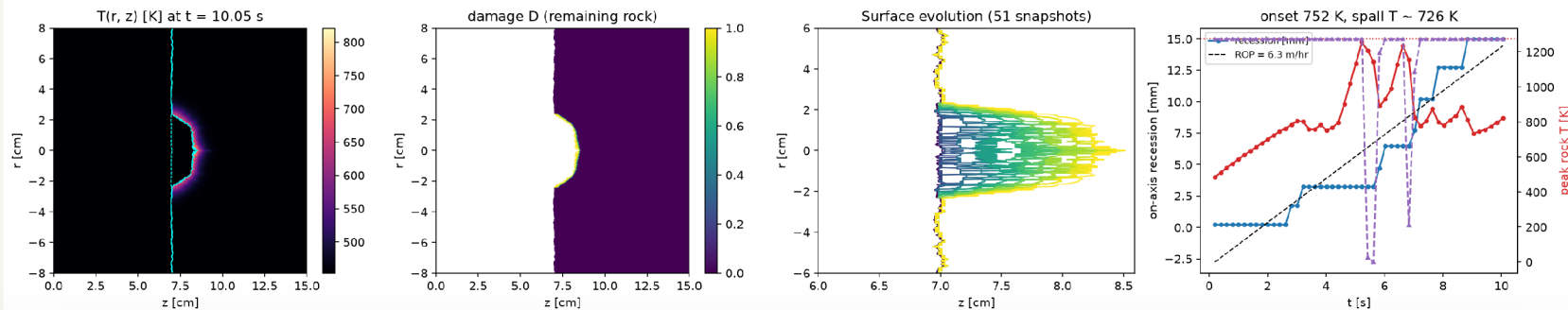
Breakdown driver vs reflected power



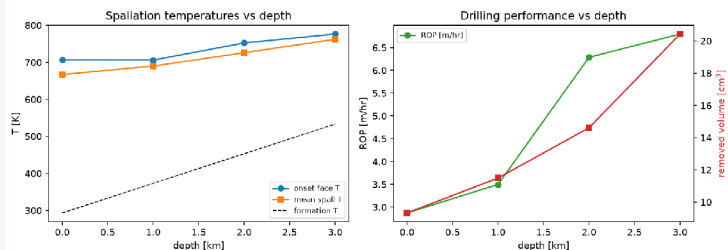
Flat-Top Beam with Thermostatic Control

$\sim 2 \text{ W/mm}^2$ ($\sim 30 \text{ kW}$ at 5.5" diameter) — power held near the melt threshold

Thermal spallation: granite, flattop, 2.5 kW

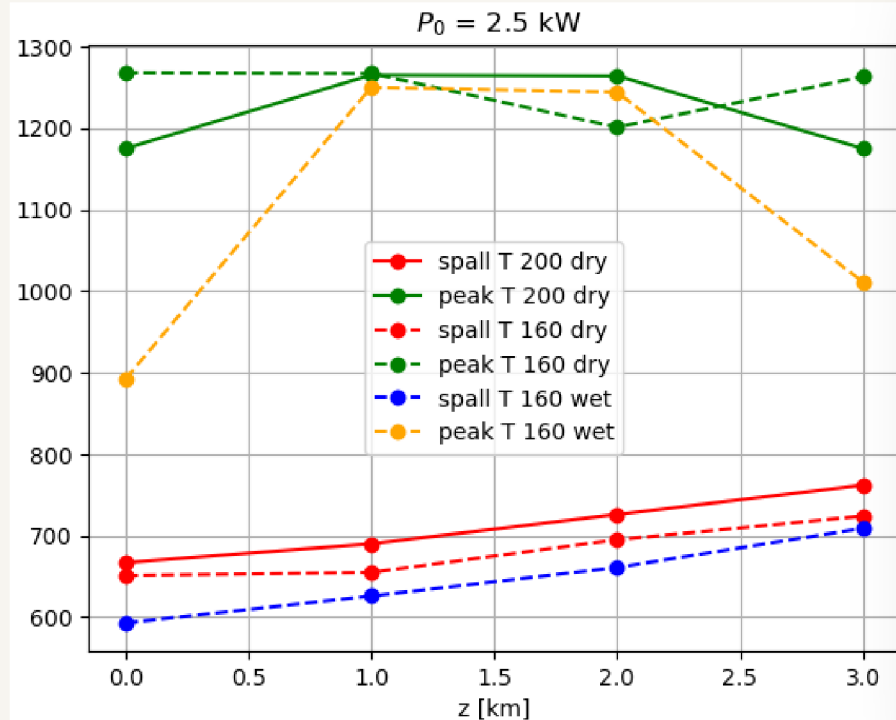


Overburden-assisted spallation: granite, flattop, 3 kW



Spallation Temperature vs. Depth

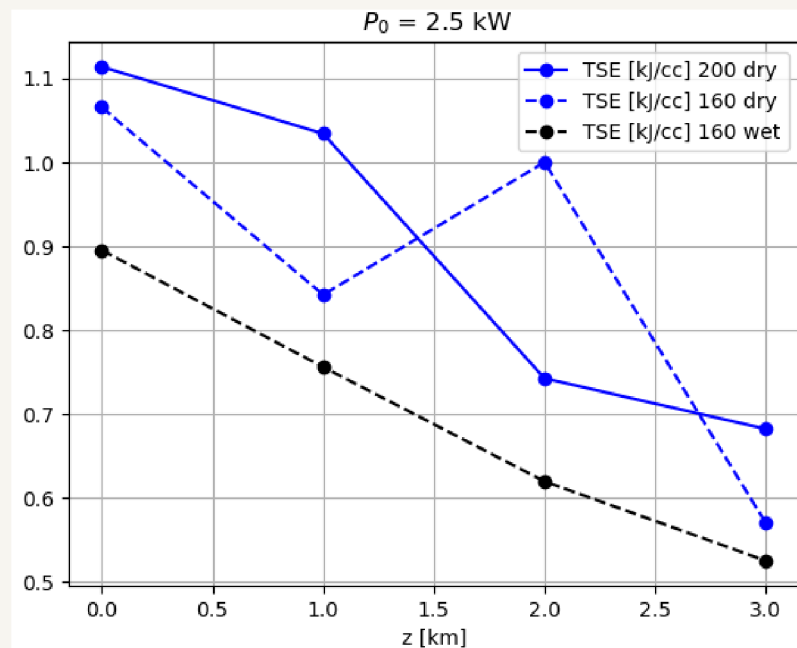
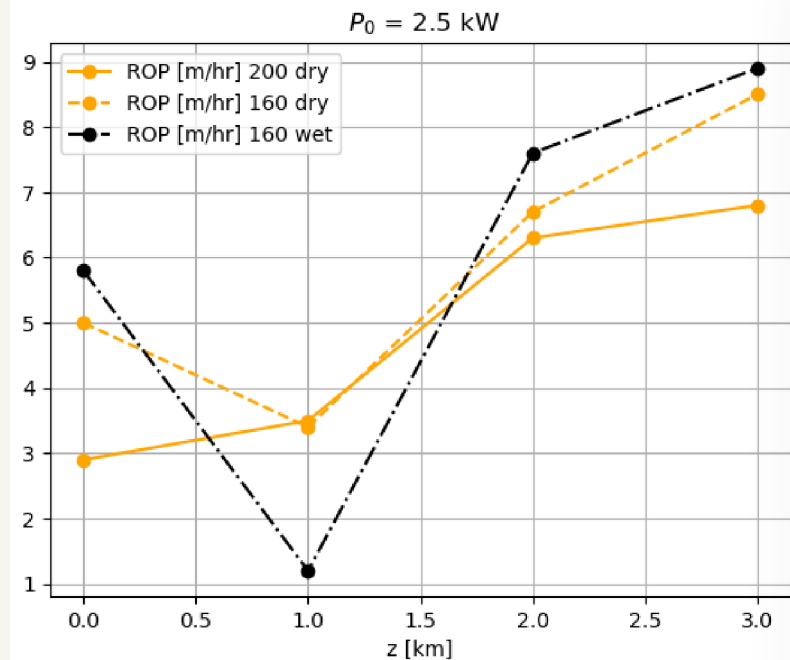
Formation gradient assumed at 80°C/km



WELLBORE STABILITY

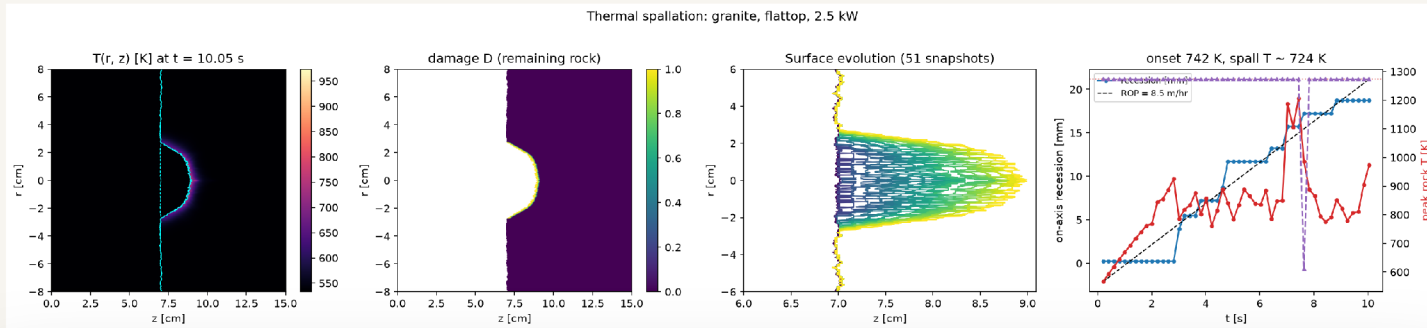
Depth	$\sigma_{ci}(T)$	Bottom	Sidewall	
1 km	159	36	57	stable
2 km	159	73	114	stable
2.5 km	158	91	143	marginal
3 km	158	109	172	sidewall breakout

Rate of Penetration and Specific Energy vs. Depth

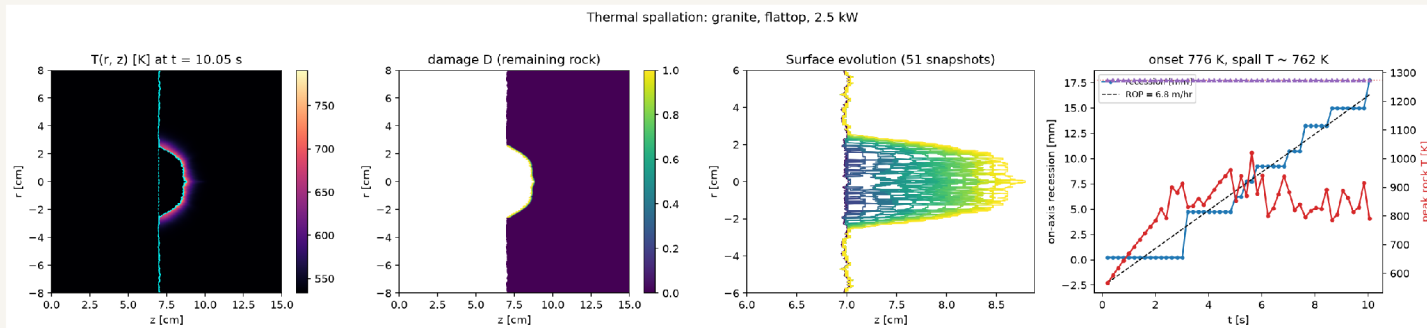


Unconfined Compressive Strength Effect at 3 km

UCS = 160
MPa, dry



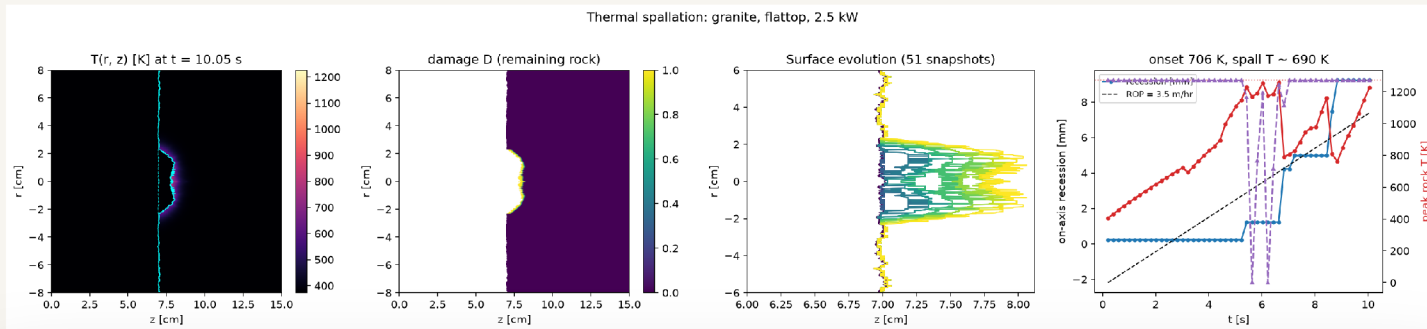
UCS = 200
MPa, dry



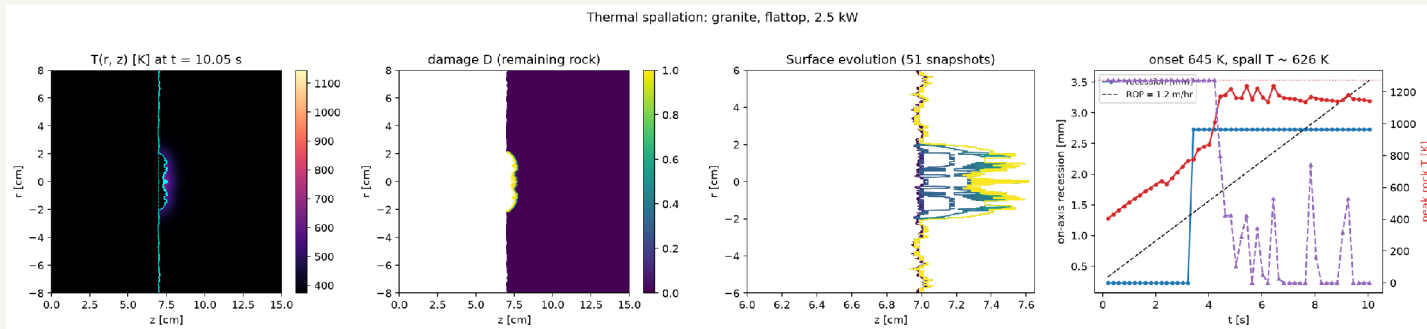
04 · DRILLING PERFORMANCE

Dry vs. Wet Granite at 1 km

Dry



Wet (0.2%
pore water)



Observations and Recommendations

OBSERVATIONS

- Beam reflections create a high-intensity region above the ablation zone
- Spall formation and expulsion via buckling gives realistic results
- Peak temperatures can reach fusion temperatures; mean spall temperature is much lower
- ROP increases with depth, with modest well pressure sensitivity
- Thermostatic power control avoids melting
- Sidewall breakout observed at 3 km
- Pore water has a significant effect up to the critical temperature

RECOMMENDATIONS

Thermal control

Measure spall size and thickness against the buckling criterion; hold surface temperature below melt with thermostatic control; hybrid drilling to remove unspalled material.

Wellbore integrity

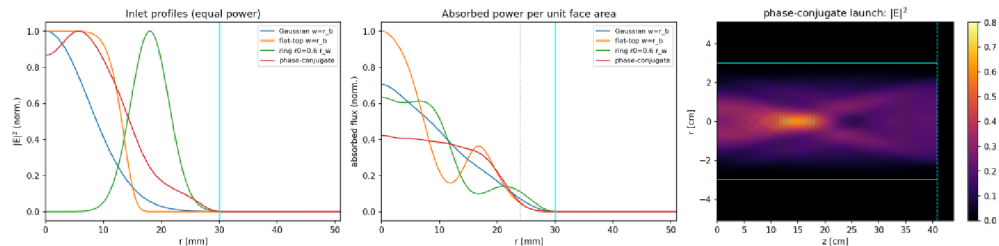
Improve wellbore cylindricity — ellipticity concentrates sidewall stress and drives breakout; ream and/or hybrid drill; run liner after 2 km with a dual string to prevent collapse.

Cuttings management

Reduce annulus size for better cutting transport; keep intensity below 2 W/mm² (93 kW at 8.5") to limit melt and cuttings mass flux at depth.

Future Work: Robust Beam Profile

Shaping the inlet for a flat hole bottom ($r_w = 30$ mm, stand-off 408 mm)



Stand-off sensitivity in a 30 mm wellbore ($L_0 = 408$ mm)

